

■ POLARTWIN — Antarctic Digital Twin Platform

Codebase Architecture, Subsystem Inventory & Lines of Code (LOC) Analysis

Ministry of Earth Sciences (MoES) • National Centre for Polar and Ocean Research (NCPOR) • SIH26060

Audit Date: September 23, 2026	Total Files: 160 source files	Total Volume: 30,001 Lines of Code	Test Status: 56/56 Passed (100%)
---------------------------------------	--------------------------------------	---	---

1. Executive Summary & Code Metrics by Layer

Layer / Subsystem	Files	LOC	% Code	Core Technologies & Architectural Role
Backend Core & Physics Engine	71	8,410	28.0%	Python 3.11, FastAPI, SQLAlchemy 2.0, SimPy, Scikit-Learn (16 physical domains, causal propagation)
Backend Automated Test Suite	10	516	1.7%	Pytest, Pytest-Asyncio, HTTPX (Unit, integration, blizzard scenario tests; 56/56 passing)
Frontend Presentation & 3D Twin	68	20,617	68.7%	React 18, TypeScript, Three.js / React Three Fiber, Apache ECharts, Tailwind CSS, Zustand
Root Configs & Documentation	11	458	1.5%	Markdown documentation, SIH matrix, environment templates, deployment configs
TOTAL REPOSITORY SUMMARY	160	30,001	100.0%	Complete, production-ready Antarctic Digital Twin Platform

2. Backend Breakdown (Python / FastAPI / SQLAlchemy) — 8,926 LOC

2.1 Application Core & Startup (`backend/app/`)

File Path	LOC	Role & Description
backend/app/main.py	232	FastAPI application entry point, CORS middleware, lifespan background tick engine startup, API router registration.
backend/app/config.py	74	Pydantic Settings configuration (DATABASE_URL, JWT secrets, simulation parameters, telemetry tick intervals).
backend/app/database.py	19	SQLAlchemy asynchronous/synchronous engine and SessionLocal factory with SQLite/Postgres support.
backend/app/deps.py	56	FastAPI dependency injection handlers for DB sessions, JWT token validation, and current user retrieval.
backend/app/seed.py	91	Seed database with initial assets, station profiles (Maitri & Bharati), default users, and initial inventory.

2.2 Database ORM Models (`backend/app/models/`)

File Path	LOC	Role & Description
backend/app/models/station.py	34	Station (name, coordinates, location type) and StationAsset (3D positions, status, metadata).
backend/app/models/alerts.py	27	Alert (severity, domain, trigger source, resolved status, timestamp).
backend/app/models/personnel.py	18	Personnel (role, station assignment, medical status, vitals, work shifts).
backend/app/models/resources.py	18	Resource (fuel, water, food, spares inventory levels and reorder thresholds).
backend/app/models/scenarios.py	18	WhatIfScenario (simulation overrides, created by, executed state).

backend/app/models/forecasts.py	17	TelemetryForecast (forecast horizon, confidence bounds, predicted values).
backend/app/models/telemetry.py	14	TelemetrySnapshot (historical domain metrics snapshot for audit & training).
backend/app/models/maintenance.py	11	MaintenanceRecord (work orders, MTBF degradation metrics, repair history).
backend/app/models/user.py	9	User (email, hashed password, role: Admin, Operator, Viewer).

2.3 REST API Endpoints (`backend/app/routers/`)

File Path	LOC	Role & Description
backend/app/routers/resources.py	215	CRUD & tracking for fuel reserves, water tanks, food cold storage, and spare parts.
backend/app/routers/analytics.py	158	Cross-domain correlation matrix, subsystem efficiency scores, and historical rollups.
backend/app/routers/telemetry.py	154	Live snapshot fetch, domain-specific telemetry streams, historical buffer queries.
backend/app/routers/scenarios.py	87	What-If simulation execution, scenario persistence, and delta comparative runs.
backend/app/routers/alerts.py	83	Alert retrieval, acknowledgment, resolution, and risk threshold adjustments.
backend/app/routers/stations.py	62	Station metadata, geographical coordinates, physical constants, asset spatial mappings.
backend/app/routers/equipment.py	48	Critical machinery statuses, generator load metrics, vibration sensors.
backend/app/routers/admin.py	37	Administrative controls, user provisioning, database re-seeding triggers.
backend/app/routers/monitoring.py	37	Subsystem health diagnostics, telemetry tick heartbeats, and memory metrics.
backend/app/routers/auth.py	32	OAuth2 password flow, JWT generation, login verification.
backend/app/routers/risk.py	24	Real-time composite risk index evaluation (0–100) per station.
backend/app/routers/forecast.py	17	ML-driven 24h/7d telemetry forecasts for energy, temperature, and fuel.
backend/app/routers/anomalies.py	15	Anomaly detection alerts triggered by IsolationForest / Z-Score engines.
backend/app/routers/recommendations.py	15	Decision intelligence and prescriptive action recommendations.
backend/app/routers/optimization.py	12	RL-based resource dispatch and load shedding optimization vectors.

2.4 Deterministic Simulation & 16 Physical Domains (`backend/app/simulation/`)

File Path	LOC	Role & Description
backend/app/simulation/inventory.py	783	Warehouse logistics, spare parts depletion curves, shelf-life and degradation modeling under polar cold.
backend/app/simulation/communication.py	705	Satellite pass windows, radome heating against blizzards, Inmarsat/LEO SNR and atmospheric attenuation.
backend/app/simulation/personnel.py	691	Crew circadian cycles, thermal stress, oxygen/calorie burn rates, medical illness probabilities.
backend/app/simulation/equipment_sim.py	603	Generator run hours, mechanical vibration, thermal expansion degradation, and Weibull reliability curves.
backend/app/simulation/fuel.py	550	Multi-tank Arctic diesel inventory, burn rates as a function of electrical load, gel-point freezing curves.
backend/app/simulation/logistics.py	514	Traverse convoy logistics, snowcat/PistenBully fuel burn, sea-ice thickness, shipping window constraints.
backend/app/simulation/water.py	406	Priyadarshini Lake vs. Seawater RO desalination, pipe freeze risk, heat-trace electrical demand.
backend/app/simulation/engine.py	158	Central tick loop coordinator; executes causal graph propagation across all 16 domains.
backend/app/simulation/environment.py	133	Katabatic wind dynamics, solar zenith angle, temperature chill factor, blizzard severity index.
backend/app/simulation/simpy_engine.py	94	SimPy discrete-event simulation wrapper for queue-based resource contention.
backend/app/simulation/energy.py	86	Microgrid balance: Solar PV + CHP + Diesel genset dispatch, bus voltage, battery state of charge.

backend/app/simulation/station_ops.py	69	Composite readiness score calculation across all subsystems (0–100%).
backend/app/simulation/maintenance_sim.py	43	Preventive maintenance scheduling and post-repair health resets.
backend/app/simulation/infrastructure.py	38	Habitat structural wind stress, envelope thermal resistance (R-value), airlock seal integrity.
backend/app/simulation/waste.py	37	Incinerator emissions, biological wastewater treatment, Madrid Protocol compliance tracking.
backend/app/simulation/safety.py	35	Fire suppression pressure, gas leak hazards, blizzard lockdown readiness.
backend/app/simulation/research.py	34	Scientific laboratory electrical draw (spectrometers, seismographs, auroral cameras).
backend/app/simulation/supplies.py	30	Food pantry inventories, blast freezer integrity (-20°C), emergency iron rations.

2.5 Decision Intelligence & AI/ML (backend/app/intelligence/)

File Path	LOC	Role & Description
backend/app/intelligence/whatif_engine.py	258	Non-destructive branching engine; clones station state and runs forward simulation under hypothetical stresses.
backend/app/intelligence/forecasting.py	249	Holt-Winters exponential smoothing + Random Forest regressors for fuel/power/temperature horizons.
backend/app/intelligence/shap_engine.py	156	SHAP feature attribution calculator for explaining causal risk scores to station commanders.
backend/app/intelligence/anomaly.py	126	Multivariant IsolationForest + rolling Z-score outlier detection on high-frequency sensor streams.
backend/app/intelligence/monte_carlo.py	114	Probabilistic blizzard survival risk analysis via 1,000+ stochastic perturbation runs.
backend/app/intelligence/rl_optimizer.py	102	Q-learning / heuristic reinforcement optimizer for automated load-shedding and battery management.
backend/app/intelligence/predictive_maintenance.py	90	Remaining Useful Life (RUL) estimation using sensor vibration and thermal variance histories.
backend/app/intelligence/risk_engine.py	82	Multi-factor weighted index aggregating life support, fuel margin, and weather severity.
backend/app/intelligence/recommendations.py	57	Rule- and model-driven prescriptive actions (e.g. 'Engage heat-trace circuit 4B to prevent freeze').
backend/app/intelligence/explainability.py	24	Natural language synthesis of causal propagation chains for operational alerts.

2.6 Services, Schemas, Security & WebSockets (backend/app/)

File Path	LOC	Role & Description
backend/app/services/weather_service.py	199	Live meteorological data provider with synthetic fallback modeling realistic Antarctic patterns.
backend/app/schemas/all_schemas.py	102	Complete Pydantic schemas validating all incoming/outgoing REST and WebSocket payloads.
backend/app/websocket/manager.py	44	Connection pooling and high-frequency JSON broadcasting over WebSocket channels.
backend/app/security/auth.py	40	Bcrypt password hashing and JWT token creation/verification.
backend/app/security/rbac.py	16	Role-Based Access Control guards (admin, operator, viewer).

2.7 Backend Automated Test Suite (backend/tests/) — 56/56 Passed

File Path	LOC	Role & Description
backend/tests/unit/test_personnel_simulation.py	161	Validates crew calorie, water, oxygen consumption, and thermal fatigue.

backend/tests/unit/test_communication_simulation.py	140	Validates satellite tracking windows, atmospheric noise, and radome heating.
backend/tests/unit/test_fuel_simulation.py	56	Validates fuel burn formulas, gel-point freezing curves, and tank balance.
backend/tests/unit/test_water_simulation.py	52	Validates freshwater lake pumping, RO energy consumption, and heat-trace logic.
backend/tests/unit/test_intelligence_advanced.py	51	Validates Holt-Winters forecasting, Monte Carlo distributions, and RL actions.
backend/tests/integration/test_api_telemetry.py	38	End-to-end integration tests for telemetry endpoints and auth tokens.
backend/tests/scenario/test_storm_scenario.py	34	Simulates a 72-hour severe blizzard cascade failure event.
backend/tests/unit/test_risk_engine.py	22	Validates composite risk scoring and boundary condition penalties.
backend/tests/unit/test_anomaly.py	11	Tests IsolationForest anomaly flag triggering on anomalous sensor readings.

3. Frontend Breakdown (React 18 / TypeScript / Three.js) — 20,617 LOC

3.1 Pages & Views (frontend/src/pages/) — 9,737 LOC

File Path	LOC	Role & Description
frontend/src/pages/PersonnelPage.tsx	1266	Expedition roster, vitals monitoring, thermal exposure tracker, shift schedules.
frontend/src/pages/CommunicationPage.tsx	850	Satellite sky orbit tracking, RF signal strength, antenna de-icing controls.
frontend/src/pages/FuelPage.tsx	781	Tank manifold telemetry, fuel gelation margins, diesel burn rate dials.
frontend/src/pages/EnvironmentPage.tsx	741	Wind compass, ambient temperature charts, blizzard index, solar irradiance.
frontend/src/pages/DecisionIntelligencePage.tsx	716	AI forecasting cockpit, SHAP causal explainability charts, RL recommendation cards.
frontend/src/pages/WaterPage.tsx	660	Priyadarshini Lake intake / Seawater RO plant telemetry, pipe freeze risk meters.
frontend/src/pages/LogisticsPage.tsx	610	Convoy traverses, sea-ice thickness radars, supply shipment schedules.
frontend/src/pages/EquipmentPage.tsx	605	Generator vibration analysis, cumulative run hours, condition-based work orders.
frontend/src/pages/InventoryPage.tsx	573	Critical spares catalog, lubricant levels, warehouse storage conditions.
frontend/src/pages/DomainsPage.tsx	502	Comprehensive multi-domain dashboard with comparison modals and drill-downs.
frontend/src/pages/Twin3DPage.tsx	387	Full-screen interactive 3D Antarctic digital twin container with telemetry overlays.
frontend/src/pages/AnalyticsPage.tsx	366	Cross-subsystem correlation heatmaps, efficiency trends, energy balance.
frontend/src/pages/ResourceMonitoringPage.tsx	361	Unified resource consumption view (power, water, fuel, oxygen, rations).
frontend/src/pages/OptimizationPage.tsx	359	Reinforcement learning resource allocation optimizer and what-if adjustments.
frontend/src/pages/WhatIfPage.tsx	353	Interactive parameter perturbation sandbox (ambient temp drop, generator outage).
frontend/src/pages/RiskAlertsPage.tsx	295	Active alarms, incident timeline, risk score history, acknowledgment workflows.
frontend/src/pages/DashboardPage.tsx	280	Executive station overview, high-level status cards, and primary dials.
frontend/src/pages/ForecastPage.tsx	236	24-hour and 7-day predictive telemetry charts with confidence bands.
frontend/src/pages/StationTwinPage.tsx	227	Station cockpit with health gauge, KPI sparklines, and interactive topology.
frontend/src/pages/LoginPage.tsx	197	Polar dark authentication portal with persona quick-selector buttons.
frontend/src/pages/AdminPage.tsx	156	System administrative console, user list, simulation speed controls.
frontend/src/pages/RecommendationsPage.tsx	135	Prescriptive action center with expected risk impact assessments.

3.2 Dashboard & Specialized UI Components (`frontend/src/components/dashboard/`) — 3,467 LOC

File Path	LOC	Role & Description
frontend/src/components/dashboard/StationFlowTopology.tsx	1179	Interactive SVG/canvas pipeline showing physical inter-station flows (energy, water, fuel).
frontend/src/components/dashboard/CrossDomainCausalTree.tsx	672	2D/3D causal tree displaying deterministic propagation across domains with active pulse links.
frontend/src/components/dashboard/OperationalDomainCard.tsx	521	Individual operational cockpit cards with telemetry values, status indicators, and sub-dials.
frontend/src/components/dashboard/DomainTelemetryInspectorModal.tsx	449	Detailed pop-up inspector displaying raw live telemetry attributes for any selected domain.
frontend/src/components/dashboard/DomainSummaryGrid.tsx	236	Responsive grid layout organizing the 16 operational domains.
frontend/src/components/dashboard/MiniDomainInstruments.tsx	232	Compact micro-dials and telemetry chips for secondary metrics.
frontend/src/components/dashboard/StationCard.tsx	141	High-level station summary card with location, coordinates, and readiness score.
frontend/src/components/dashboard/StationHealthGauge.tsx	67	Circular SVG readiness score gauge with smooth transition animations.
frontend/src/components/dashboard/GlobalAlertBanner.tsx	41	Top-level alert ticker alerting operators to active critical incidents.

3.3 3D Spatial Digital Twin (`frontend/src/components/twin3d/`) — 1,743 LOC

File Path	LOC	Role & Description
frontend/src/components/twin3d/StationScene.tsx	1546	Three.js / React Three Fiber Antarctic station 3D digital twin: terrain, station buildings, gensets, fuel tanks, water pipes, snow particle effects, and spatial asset clicking.
frontend/src/components/twin3d/AssetInfoPanel.tsx	197	Floating glassmorphism HUD displaying real-time telemetry of 3D objects clicked in the scene.

3.4 Charts & Visualization Engines (`frontend/src/components/charts/`) — 2,248 LOC

File Path	LOC	Role & Description
frontend/src/components/charts/CausalGraphViewer.tsx	920	Full-featured node-link causal graph visualizer with layout algorithms and interactive filters.
frontend/src/components/charts/WindCompass.tsx	295	Custom SVG Antarctic katabatic wind compass showing live speed, bearing, and gust vectors.
frontend/src/components/charts/EChartsLine.tsx	218	High-performance Apache ECharts line graph for telemetry histories and confidence intervals.
frontend/src/components/charts/PolarEChartsAnalytics.tsx	212	Polar coordinate charts for multi-dimensional telemetry comparison.
frontend/src/components/charts/SparklineChart.tsx	144	Compact SVG trend sparklines embedded directly into cards and table rows.
frontend/src/components/charts/IndustrialGauge.tsx	142	High-contrast industrial dial meters with colored threshold arcs.
frontend/src/components/charts/TankLevelBar.tsx	113	Liquid reservoir visualization with animated surface levels and percentage readouts.
frontend/src/components/charts/RadarRiskChart.tsx	108	Multi-axis radar chart showing vulnerability across subsystems.
frontend/src/components/charts/RealTimeAlertFeed.tsx	97	Live scrolling list of telemetry threshold excursions.
frontend/src/components/charts/EChartsBar.tsx	91	Bar charts for categorical comparisons (e.g. inventory levels, energy usage by lab).

3.5 State Management & Data Layer (`frontend/src/store/`, `api/`, `hooks/`, `types/`)` — 1,607 LOC

File Path	LOC	Role & Description
<code>frontend/src/types/index.ts</code>	422	Single source of truth for TypeScript interfaces covering all 16 domains, alerts, and assets.
<code>frontend/src/utils/domainDataHelper.ts</code>	322	Formatting utilities, unit conversions, domain color definitions, and label mappings.
<code>frontend/src/store/telemetryStore.ts</code>	242	Primary Zustand store holding live snapshots, historical buffers, and risk metrics.
<code>frontend/src/api/client.ts</code>	242	Central Axios HTTP client with JWT interceptors, base URL config, and type-safe API calls.
<code>frontend/src/hooks/useWebSocket.ts</code>	101	Custom React hook managing real-time WebSocket connection, reconnection logic, and dispatching to Zustand.
<code>frontend/src/store/authStore.ts</code>	44	Zustand store managing JWT token lifecycle, user credentials, and role state.
<code>frontend/src/store/alertStore.ts</code>	41	Zustand store managing station alerts, filter states, and acknowledgment counts.
<code>frontend/src/store/scenarioStore.ts</code>	41	Zustand store holding active What-If simulation parameters and results.
<code>frontend/src/store/stationStore.ts</code>	34	Station selector store managing active station focus (maitri vs bharati).
<code>frontend/src/store/uiStore.ts</code>	11	Minor UI state (modals, drawer toggles).

3.6 Layout, Design System & Configs (`frontend/src/components/common/`, Root Frontend`)` — 764 LOC

File Path	LOC	Role & Description
<code>frontend/src/components/common/Navbar.tsx</code>	260	Header navigation, station switcher, composite risk status badge, user logout.
<code>frontend/src/index.css</code>	165	Polar dark design system CSS variables, glassmorphism utilities, scrollbars, and keyframe animations.
<code>frontend/src/App.tsx</code>	97	React Router DOM configuration, protected route layout wrapping, and route declarations.
<code>frontend/src/components/common/Layout.tsx</code>	64	Main dashboard shell housing the Navbar and content viewport with background weather sync.
<code>frontend/tailwind.config.js</code>	46	Tailwind tokens extending custom polar colors, fonts (Inter, JetBrains Mono), and glass shadows.
<code>frontend/package.json</code>	41	Frontend dependencies (React, Three.js, Lucide, ECharts, Zustand, Tailwind).
<code>frontend/src/components/common/ProtectedRoute.tsx</code>	38	Route guard redirecting unauthenticated users to /login.
<code>frontend/src/main.tsx</code>	9	React DOM root mounting and strict mode wrapping.

4. Root Project Files & Documentation — 458 LOC

File Path	LOC	Role & Description
<code>README.md</code>	297	Full architectural documentation, hackathon details, quick start instructions, and feature guide.
<code>docs/IMPLEMENTATION_MATRIX.md</code>	76	Requirements coverage matrix for SIH26060 across the 16 operational domains.
<code>.env.example</code>	35	Environment variable template for database, JWT, and simulation timing.
<code>.env</code>	29	Active local development environment variables.
<code>.gitignore</code>	21	Git ignore rules for node_modules, python venv, database files, and caches.

5. Runtime Operational Status & Readiness

- **Backend (FastAPI):** Running live on `http://127.0.0.1:8000` with Swagger interactive docs at `http://127.0.0.1:8000/docs`.
- **Frontend (React / Vite):** Running live on `http://localhost:5173/` with fast HMR.

- **Automated Tests:** 100% test pass rate across 56 test modules (pytest tests -q).
- **Theme:** Locked permanently to the high-contrast Antarctic Polar Dark theme.
- **Status:** Standing by for operator input and directives before any modifications or enhancements.